

Key Words: Antibiotics, penicillin, drugs, Alexander Fleming, digitalis, foxglove, willow, aspirin

Starter: Answer these three review questions below.

1. Name an antibiotic and what it does. [2 marks]

2. Antibiotics can not kill _____
because they live and _____
_____ of cells. [3]

3. Describe three advantages
of antibiotics [4]



Antibiotics

Review

1. Name an antibiotic and what it does. [2 marks]

Penicillin [1]

Damages bacterial cells [1]

2. Antibiotics can not kill viruses because they live and reproduce inside of cells. [3]

3. Describe three advantages of antibiotics [4]

Damage bacterial cells without damaging Human cells [1]

Easy to administer by tablet or liquid [1]

Saves many people's lives [1]

Key Words: Antibiotics, penicillin, drugs, Alexander Fleming, digitalis, foxglove, willow, aspirin

Starter: Let's do a quiz on what we learnt last lesson. 1 □ 10



Quiz:

1. What is an antibiotic?
2. State an example.
3. Why do antibiotics not destroy viruses?
4. State a cause for concern with antibiotics.
5. What is MRSA?
6. Name two first lines of defense against pathogens
7. **Phagocytes** and **lymphocytes** are types of WBC. Which one releases antibodies and antitoxins?
8. Name two viral and two bacterial diseases.
9. State a difference between a bacterium and a virus.
10. What is a vector?

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Starter: Complete this retrieval triangle.

Last
lesson

Why don't antibiotics
destroy viruses?

2 lessons
ago

Use the words **inject**
antibodies, **antigens** and
memory cells to write
about vaccines

Define 'Herd
Immunity'.

1
unit
ago

Measles
symptoms and
treatment

How do you
prevent malaria
spreading

Describe
gonorrhoea.

Last
lesson

Why don't antibiotics
destroy viruses?

Viruses live inside of cells

2 lessons
ago

Use the words **inject**
antibodies, **antigens** and
memory cells to write
about vaccines

Inject weakened pathogen which cause lymphocytes to release antibodies which attach to the antigens on the Pathogen surface. Neutralises the pathogen and memory Cells are created that create antibodies faster next time.

Define 'Herd
Immunity'.

When a large amount of people in a population is immune to a disease causing the disease to reduce its spread or even disappear.

1
unit
ago

Measles
symptoms and
treatment

Symptom: Red skin
rash, fever
No treatment.
Vaccination prevents
disease.

How do you
prevent malaria
spreading

Mosquito net,
remove stagnant
water breeding sites
of mosquitos

Describe
gonorrhoea.

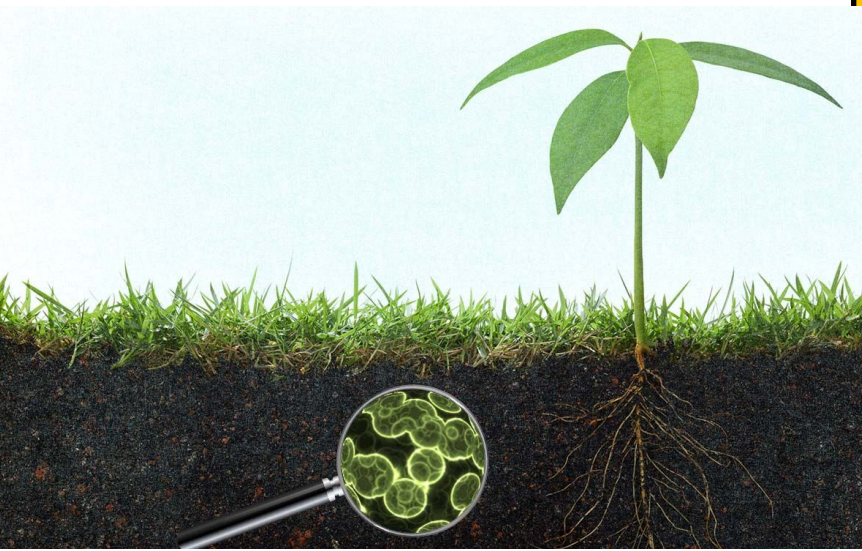
STD, bacterial
disease, causes thick
yellow/green
discharge, treated
with antibiotics

LOb: Understand how penicillin was discovered and that drugs are sourced from plants and fungi.

Keywords: Antibiotics, penicillin, drugs, Alexander Fleming, digitalis, foxglove, willow, aspirin

Learning Outcomes:

- Describe some drugs traditionally extracted from plants
- Describe and explain the discovery of penicillin
- Discover how scientists might find new drugs

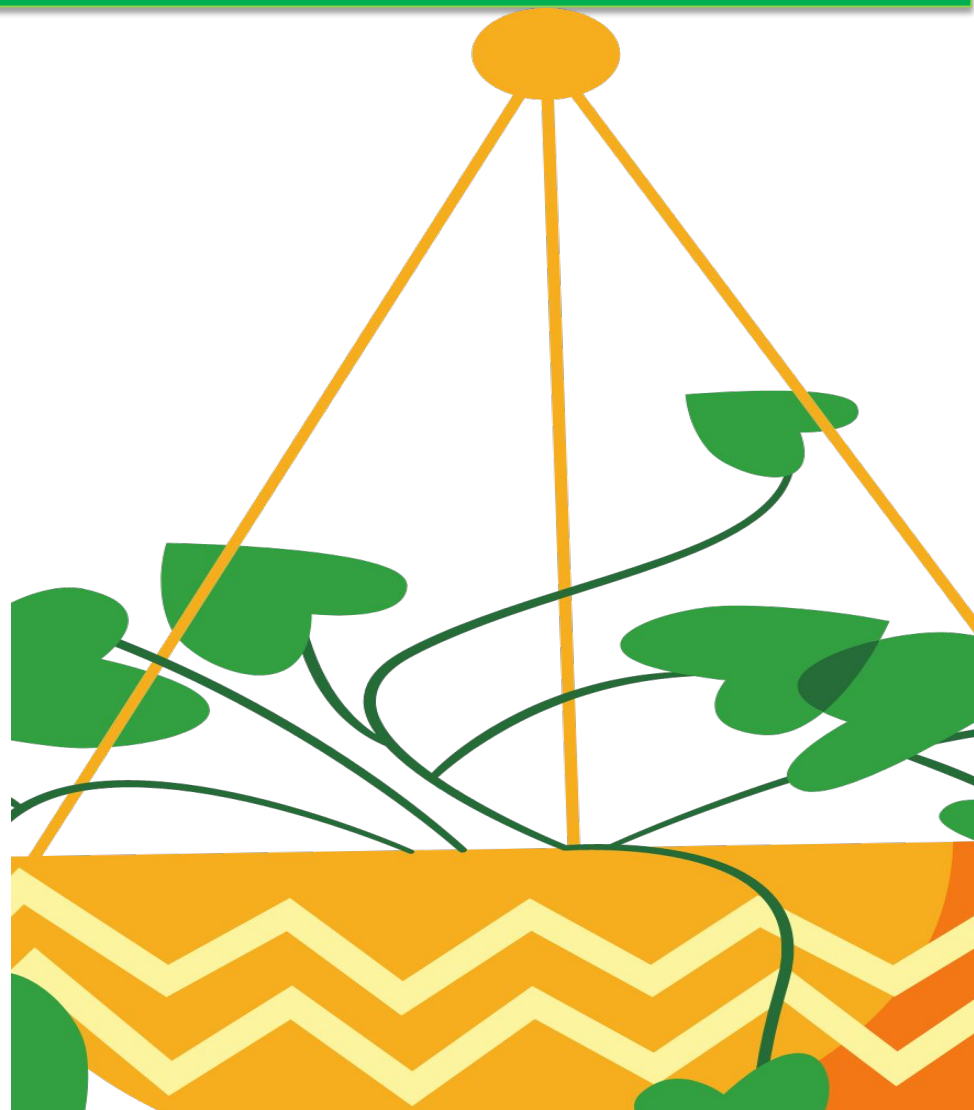
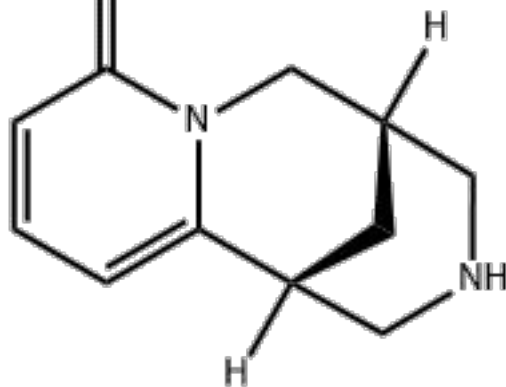


Did you know... there are around 1 billion bacteria per gram of soil extracted from the ground. They exist feeding on carbon energy sources.

There are a number of chemicals extracted from plants that are used in drugs we have developed.

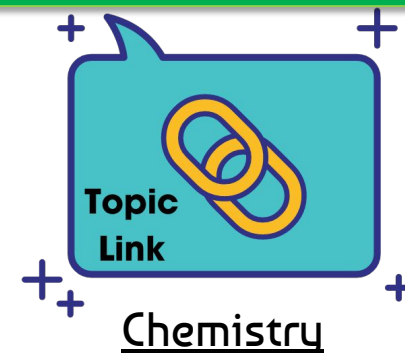
EXAMPLE

Cytisine

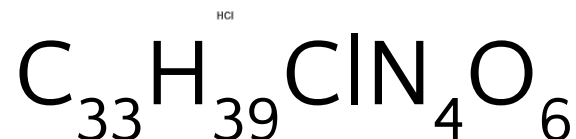
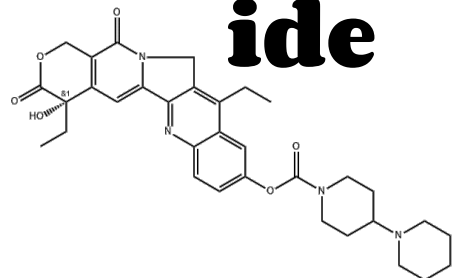


Answer these questions about this chemical

EXAMPLE



Irinotecan hydrochloride



1. How many elements in this formula?

2. How many atoms in the formula?

3. What type of bond is present between the elements in this compound?

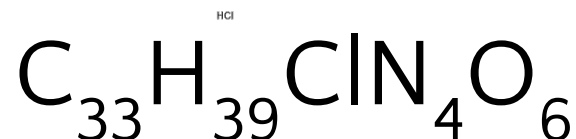
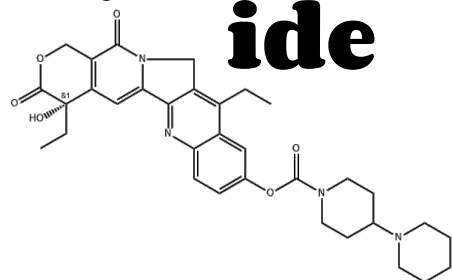
4. Define a compound.

5. Which element in the formula is a halogen

Review

EXAMPLE

Irinotecan hydrochloride



1. How many elements in this formula?

5 elements

2. How many atoms in the formula?

$33 + 39 + 1 + 4 + 6 = 83$ atoms

3. What type of bond is present between the elements in this compound?

Covalent bonds

4. Define a compound.

A substance with two or more elements chemically bonded.

5. Which element in the formula is a halogen

chlorine

There are three chemicals sourced from plants / fungi that you need to know.

**Name of
plant**

FOXGLOVE



**Name of
drug**

Digitalis and Digoxin

**Use of
drug**

Used since the 18th century and in more modern drugs to help patients with heart failure to help strengthen the heartbeat.

You need to know this!

Name of plant

Willow Tree



Name of drug

Acetyl Salicylic acid (aspirin)

Use of drug

Found in the bark of willow trees. It is a painkiller and an anti-inflammatory.

You need to know this!

Name of Fungi

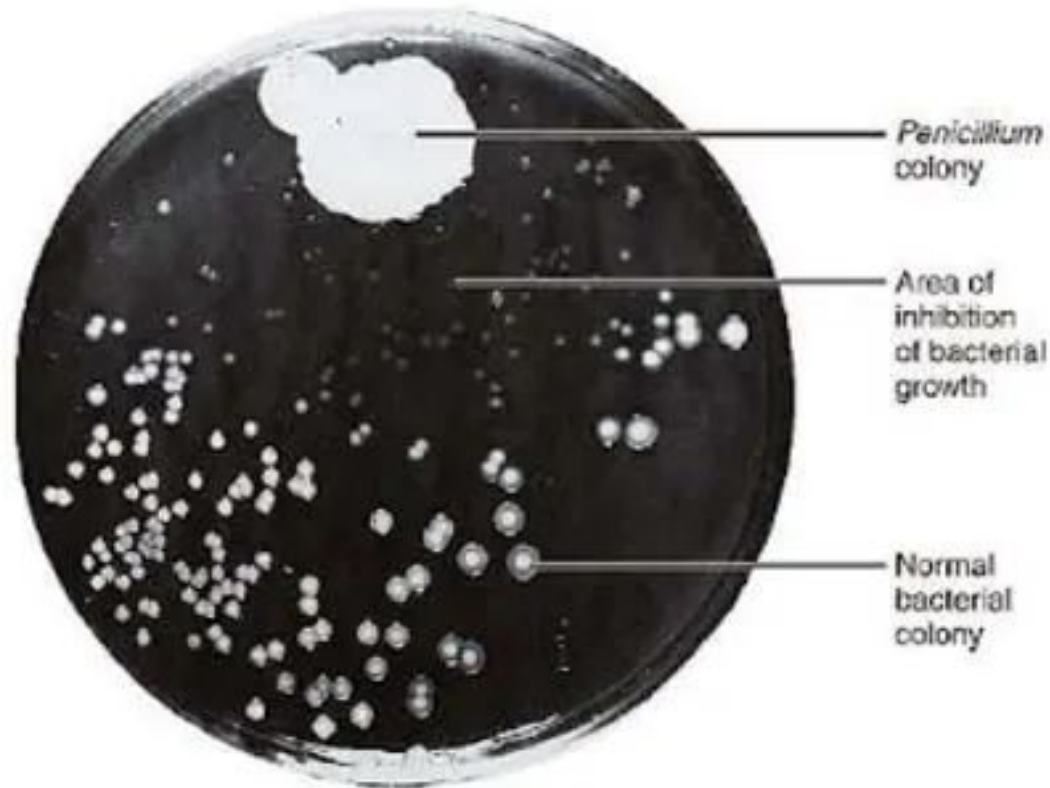
Penicillium

Name of drug

Penicillin

Use of drug

Antibiotic that kills bacteria without damaging human cells



BONUS: You don't need to know this

**Name of
plant**

Poppies



**Name of
drug**

Opium (morphine)

**Use of
drug**

Opium compounds found in poppy seeds are strong painkillers.

Review

Source of drugs



Name of drugs

A _____

P _____

D _____

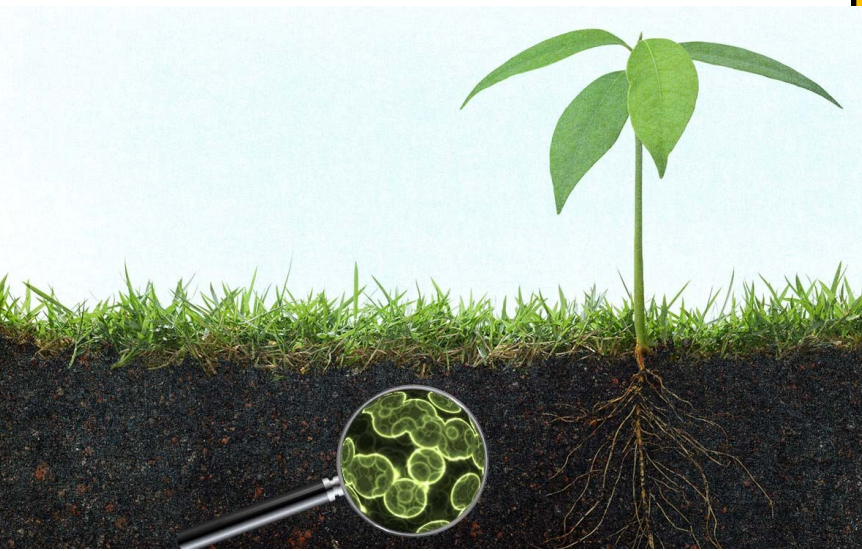
Use of drugs

LOb: Understand how penicillin was discovered and that drugs are sourced from plants and fungi.

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Did you know... there are around 1 billion bacteria per gram of soil extracted from the ground. They exist feeding on carbon energy sources.

Activity: Watch the video and assess the importance of the first antibiotic and the people who contributed to its development.

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!

In 1928, **Alexander Fleming** was growing bacteria for study purposes. He was careless, often leaving the lid open on his **agar plates**.



He went on holiday
and on his return
he saw this on his
agar plate...



He noticed a clear area where there was lack of bacterial growth. He hypothesised that the mould was producing something that killed the bacteria.

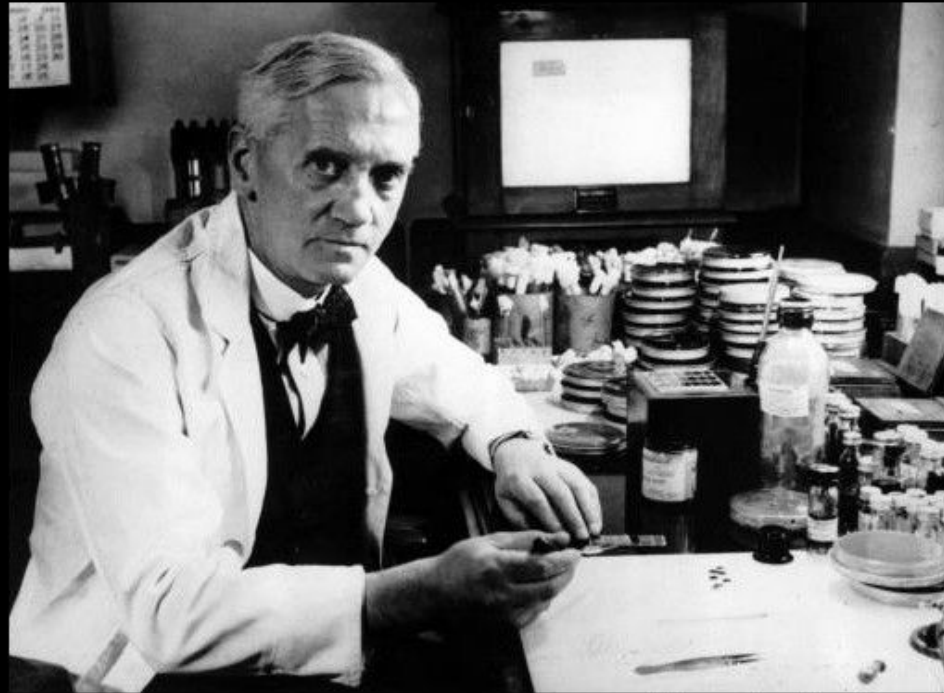


He named the substance produced '**penicillin**' after the mould 'penicillium'. He tried unsuccessfully to extract an active juice from the mould. He gave up after a while.

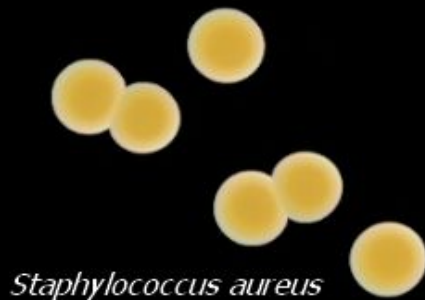
www.bacterialphotos.com



Penicillium chrysogenum
(*P. notatum*)



Alexander Fleming



Staphylococcus aureus



10 Years later these Florey and Chain tried to extract penicillin and were successful.



They tested it on a human and proved it could be of use. Drugs company 'Pfizer' mass produced the penicillin and was in use by the end of WWII.



Sir Alexander Fleming
(1881-1955)



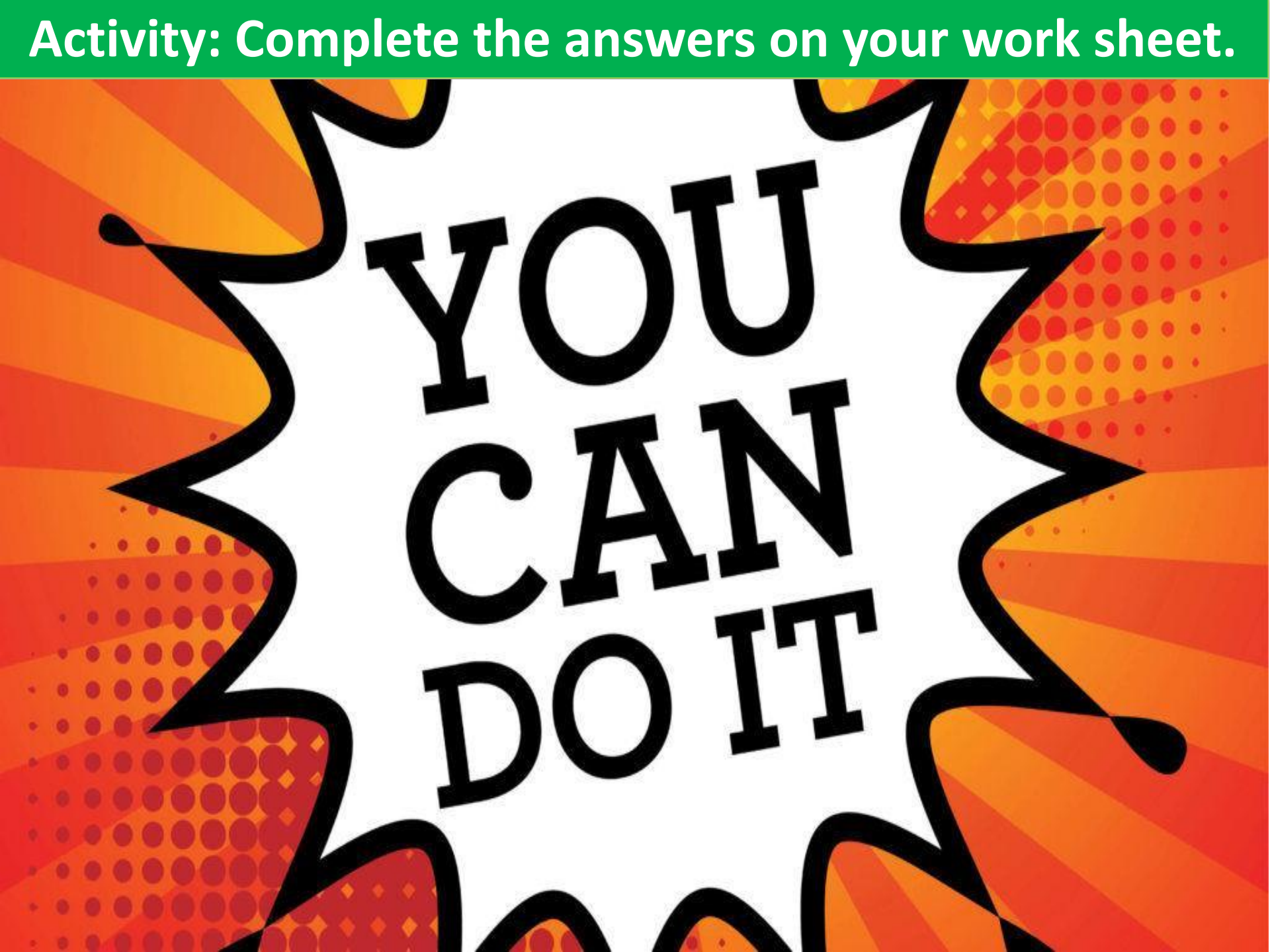
Ernst Boris Chain
(1906-1979)



Sir Howard Walter Florey
(1898-1968)

Sir Alexander Fleming discovered penicillin in 1928 and Dr Ernst Chain and Sir Howard Walter Florey scaled up production of Fleming's discovery to make medicinal quantities of penicillin. All three won the Nobel prize in 1945 for their contributions

Activity: Complete the answers on your work sheet.



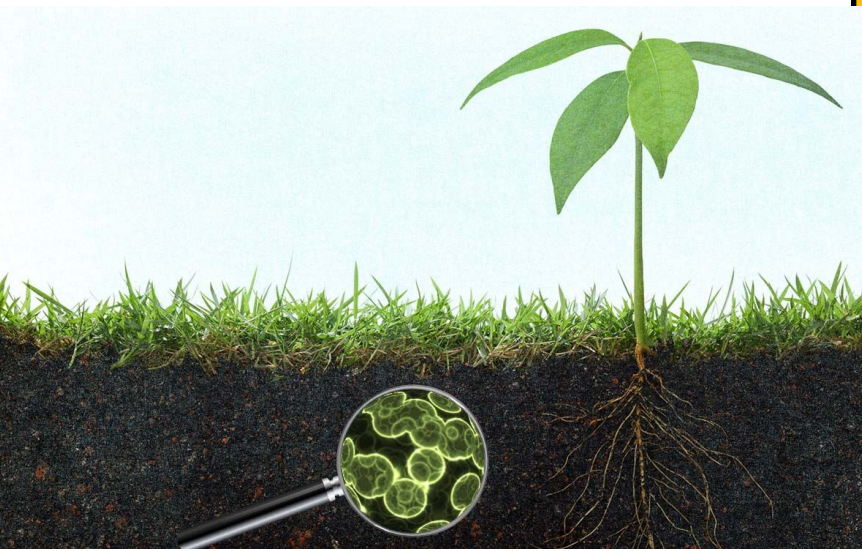
**YOU
CAN
DO IT**

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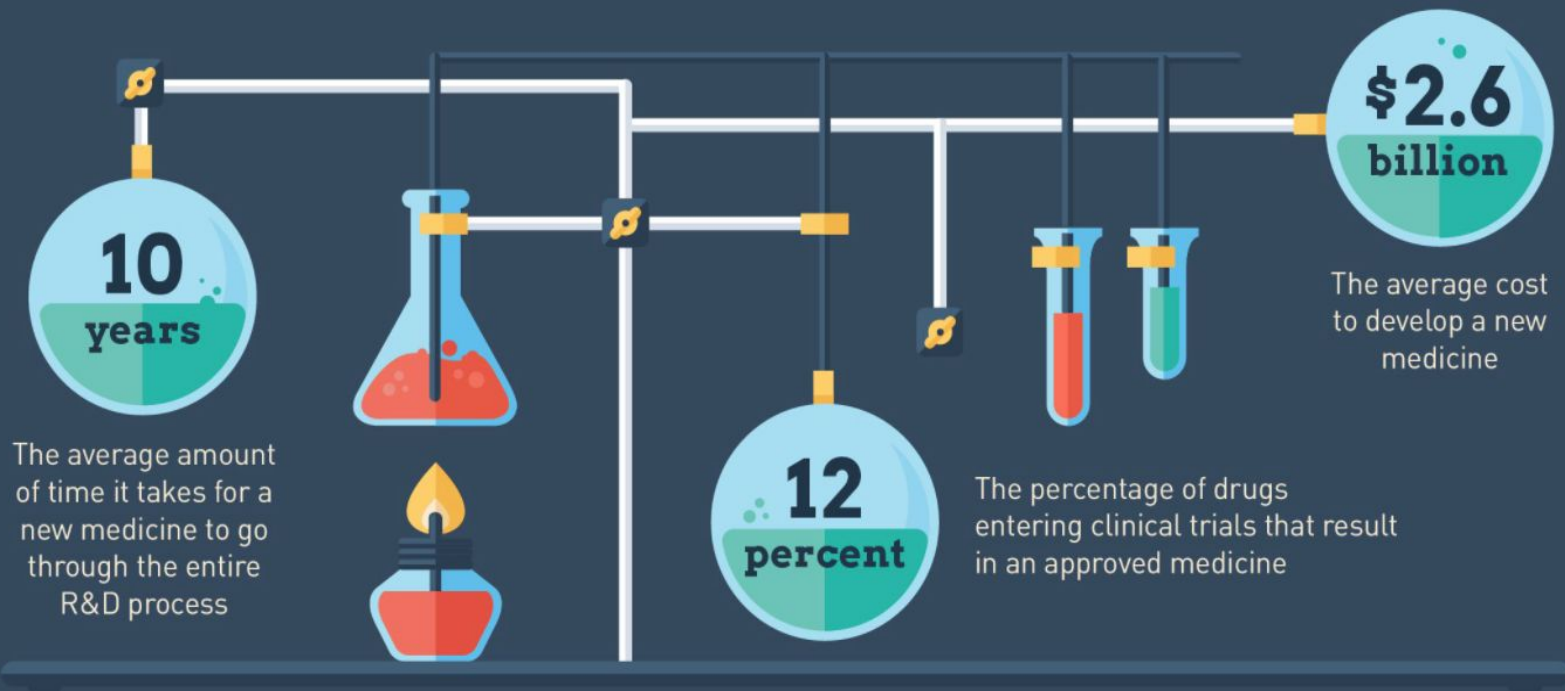


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Drug development is long, complicated and expensive.

THE CHALLENGE OF DEVELOPING NEW TREATMENTS & CURES

Innovative medicines offer great hope to patients and the health care system, but developing these new treatments and cures is a complex and risky undertaking.



Most new drugs are synthesised by chemists working in pharmaceutical companies using chemical banks and computer models.

BETWEEN 1998 AND 2014:

The number of unsuccessful vs. successful attempts to develop medicines to treat certain diseases

Unsuccessful Attempts

ALZHEIMER'S DISEASE

123



Approved Medicines

ALZHEIMER'S DISEASE

4

MELANOMA

96



MELANOMA

7

LUNG CANCER

167



LUNG CANCER

10

MORE THAN

7,000



The number of medicines currently in development around the world

70

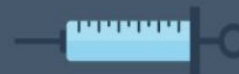
PERCENT



The percentage of new medicines in development that are potential first-in-class therapies, meaning they use a completely new approach to fighting a disease

42

PERCENT



The percentage of new medicines in the pipeline that have the potential to be personalized medicines

Plants are still a starting point for developing drugs...

**Name of
Plant**

Noni fruit

**Name of
drug**

Unknown

**Use of
drug**

Future cancer
treatments,
antibiotics



Scientists are collecting soil samples globally searching for microorganisms that produce antibiotics.

Scientists are finding new ways to grow them in a controlled way.



Did you know... only 1% of soil microbes can be grown (cultured) in the lab

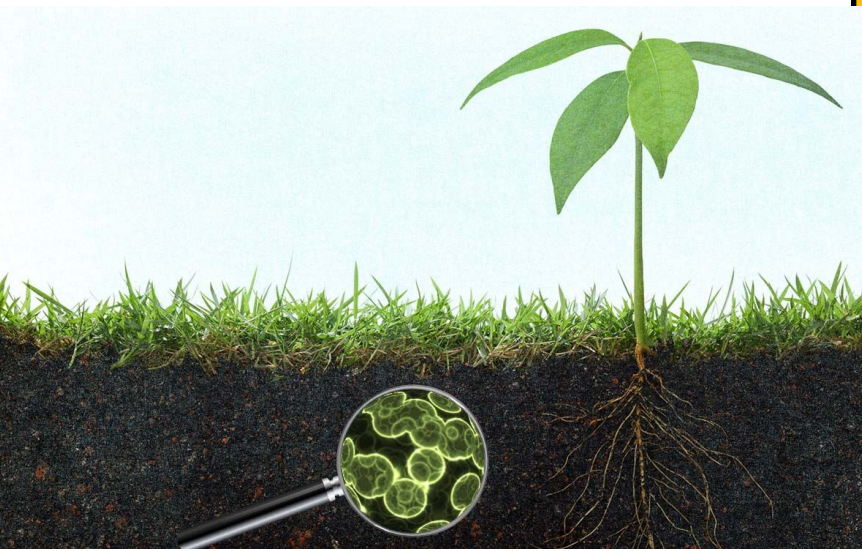
What have you learned about developing new drugs...

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keyfacts



Traditionally drugs were extracted from plants. Foxgloves = **digitalis**, willow trees = **aspirin**, fungi = **penicillin**



Penicillin was discovered by **Alexander Fleming** when he left the lid of his agar plates and **penicillium mould** grew on them.



New drugs can be **synthesised by chemists** in the **pharmaceutical industry**. The starting point may still be chemicals extracted from plants.

Plenary: Missing words

1. The _____ plant is a source of digitalis.
2. Alexander left his _____ after going on
3. ~~How~~ ~~Why~~ trees help to make _____
4. The penicillin antibiotics was named after _____.
5. New drugs are created by chemists working for _____
6. Drugs are created from chemicals extracted from _____



Plenary: Choose a Number One to Five

